Growth comparisons and the arithmetic cost of shifting odd-zeta forms

27 October 2026

Abstract

We apply polynomial and logarithmic growth comparisons to the actual primitive coefficients of tail-moment determinants for odd zeta values. An elementary prime-denominator argument gives an explicit lower bound for the primitive target coefficient, including the full gcd reduction. It proves that shifts satisfying $N^2 \log(a_N + 2N) = o(a_N)$ make the normalized height diverge rather than vanish. Thus increasing the shift sufficiently rapidly cannot supply the missing irrationality estimate. Lean verifies the growth-comparison and rational-denominator lemmas described below. The estimate required for smaller shifts remains unproved, and no all-odd irrationality conclusion is claimed.

1 The coefficients to be compared

Fix $s = 2r + 1$, $r \geq 1$, $N \geq 1$, and $a \geq 0$. Put

$$H_k^{(s)} = \sum_{j=1}^{k} j^{-s}, \qquad \Delta_{N,a} = \det[\zeta(s) - H_{a+i+j}^{(s)}]_{0 \leq i,j < N} = A_{N,a}\zeta(s) + B_{N,a}.$$

Subtracting row zero from every other row proves the affine identity. The gamma integral gives the positive weight

$$w_s(x) = \frac{(-\log x)^{s-1}}{(s-1)!(1-x)}, \qquad \zeta(s) - H_k^{(s)} = \int_0^1 x^k w_s(x)\,\mathrm{d}x.$$

In the basis $1, 1-x, x(1-x), \ldots, x^{N-2}(1-x)$, whose change-of-basis determinant is ± 1, the target coefficient is the lower Gram determinant

$$A_{N,a} = \det[(a+i+j+1)^{-s} - (a+i+j+2)^{-s}]_{0 \leq i,j < N-1}. \tag{1}$$

It is positive. Hadamard's inequality and diagonal entries at most one give $A_{N,a} \leq 1$; for $N = 1$, the empty determinant is one.

Write the rational approximation in lowest terms as

$$-\frac{B_{N,a}}{A_{N,a}} = \frac{p_{N,a}}{u_{N,a}}, \qquad p_{N,a} \in \mathbb{Z}, \quad u_{N,a} \in \mathbb{Z}_{>0}.$$

Put $Q = \operatorname{lcm}(1, \ldots, a + 2N - 2)^s$, with an empty lcm equal to one. Then $U = Q^N A$ and $V = Q^N B$ are integers. For $g = \gcd(|U|, |V|)$, we have $u = U/g$ and $p = -V/g$. Thus the following bounds concern the fully reduced coefficient, not an unnormalized presentation.

The endpoint calculation has the exact minimum

$$E_{N,a} := \frac{\Delta_{N,a}}{A_{N,a}} = \min_{\deg P < N,\ P(1)=1} \int_0^1 P(x)^2 x^a w_s(x)\,\mathrm{d}x.$$

Comparing $1-x \le -\log x \le (1-x)/x$ with the Jacobi minimum

$$\min_{\deg P<N,\ P(1)=1} \int_0^1 P^2 x^b (1-x)^m \,\mathrm{d}x = \frac{m!(m+1)!}{(N)_{m+1}(N+b)_{m+1}}$$

gives, for $N > r$,

$$\frac{(2r-1)!}{(N)_{2r}(N+a)_{2r}} \le E_{N,a} \le \frac{(2r-1)!}{(N-r)_{2r}(N-r+a)_{2r}}, \qquad (z)_k = \prod_{j=0}^{k-1}(z+j).$$

The lower comparison uses $m = 2r-1$, $b = a$; the upper one restricts $P = x^r Q$. Consequently $N^{2r}(N+a_N)^{2r}E_{N,a_N} \to (2r-1)!$ uniformly for nonnegative shifts. Small primitive forms therefore require exactly

$$\frac{u_{N,a_N}}{N^{2r}(N+a_N)^{2r}} \longrightarrow 0. \tag{2}$$

2 What growth comparison proves

If $a_N \ge 0$ and $|u_N| \le CN^p$ eventually, then

$$\left|\frac{u_N}{N^{2r}(N+a_N)^{2r}}\right| \le CN^{p-4r} \longrightarrow 0 \qquad (p < 4r).$$

More generally, a bound $|u_N| \le CN^p(\log N)^k$ with real k still suffices when $p < 4r$, since $(\log N)^k = o(N^\epsilon)$ for every $\epsilon > 0$. Lean verifies both conclusions with the numerator bounds as explicit hypotheses.

A useful general form, also verified in Lean, assumes a lower envelope $N + a_N \ge cN^\beta$, $c > 0$. Then

$$N^{2r}(N+a_N)^{2r} \ge c^{2r}N^{2r(1+\beta)},$$

so $|u_N| \le CN^p(\log N)^k$ suffices when $p < 2r(1+\beta)$. At equality a negative logarithmic exponent still suffices. This also handles fluctuating shifts that satisfy the same lower envelope. When $a_N = o(N)$, the denominator is asymptotic to N^{4r}. For a fixed negative lower bound on a_N, it is eventually at least $2^{-2r}N^{4r}$; the stronger inequality with constant one requires $a_N \ge 0$. If $a_N \asymp N^\beta$, $\beta > 1$, the denominator instead has order $N^{2r(1+\beta)}$. Every comparison must evaluate u_{N,a_N} at that same shift. A numerator estimate for $a = 0$ is not uniform in a.

A saddle-point calculation for a determinant can estimate its absolute size, but it does not estimate its coefficient gcd. Likewise a recurrence must first be proved for the reduced sequence before a recurrence asymptotic theorem can be applied to it. The next calculation exposes an arithmetic contribution that these substitutions must retain.

3 An exact centered denominator budget

Let

$$\delta_k = H_{a+k}^{(s)} - H_a^{(s)}, \qquad D = \prod_{k=1}^{2N-2}(a+k)^s, \qquad Z_{ij} = -D\delta_{i+j}.$$

An empty product is one. Each Z_{ij} is an integer, because every denominator in δ_{i+j} divides D. With J the all-ones matrix, define

$$m = \det(Z + DJ) - \det Z = D^N A_{N,a}, \qquad c = \det Z.$$

The centered determinant identity is

$$D^N \Delta_{N,a} = m(\zeta(s) - H_a^{(s)}) + c.$$

Since $0 < A_{N,a} \le 1$, m is a positive integer and $m \le D^N$. Combining this identity with the reduced rational approximation yields

$$H_a^{(s)} = \frac{p}{u} + \frac{c}{m}. \tag{3}$$

If b is the reduced denominator of $H_a^{(s)}$, then $b \mid um$: multiply (3) by um. This divisibility is unaffected by any common factor already removed from U, V.

Lemma 1. *Put*

$$T_s(a) = \prod_{\substack{p \le a,\ p \text{ prime} \\ a < 3p \\ p \nmid 2^s+1}} p.$$

Then

$$T_s(a)^s \mid b, \qquad T_s(a)^s \mid um, \qquad u \ge \frac{T_s(a)^s}{D^N}. \tag{4}$$

Proof. Fix a prime in the displayed product. If $a < 2p$, the unique summand whose denominator is divisible by p is p^{-s}. All other summands have denominators prime to p. Hence the reduced denominator contains p^s.

If $2p \le a < 3p$, isolate the two summands

$$\frac{1}{p^s} + \frac{1}{(2p)^s} = \frac{2^s+1}{2^s p^s}.$$

Their reduced denominator contains p^s, since $p \nmid 2^s + 1$. Adding a rational with denominator prime to p cannot remove that prime power: if $q - R$ has denominator divisible by p^s and $p \nmid \operatorname{den}(R)$, the divisibility $\operatorname{den}(q-R) \mid \operatorname{den}(q)\operatorname{den}(R)$ forces $p^s \mid \operatorname{den}(q)$. Distinct prime powers are coprime, so their product divides b. Now use $b \mid um$, $0 < m \le D^N$, and positivity. □

4 A proved obstruction to rapidly growing shifts

Write $\vartheta(x) = \sum_{p \le x} \log p$. The exceptional primes $p \mid 2^s + 1$ have product at most $2^s + 1$. Therefore

$$\log T_s(a) \ge \vartheta(a) - \vartheta(a/3) - \log(2^s + 1).$$

The elementary Chebyshev inequalities, also proved in the pinned mathlib source [1], are

$$\vartheta(a) \ge a \log 2 - \log(a+1) - 2\sqrt{a}\log a, \qquad \vartheta(x) \le x \log 4 \quad (x \ge 0).$$

They give the explicit estimate

$$\boxed{\log u_{N,a} \ge s\left[\frac{a\log 2}{3} - \log(a+1) - 2\sqrt{a}\log a - \log(2^s+1) - N(2N-2)\log(a+2N)\right].} \tag{5}$$

Here we used $D^N \le (a+2N)^{sN(2N-2)}$. No prime number theorem is needed.

Theorem 2. *For a fixed odd weight $s = 2r+1 > 1$, suppose $a_N \ge 1$ and*

$$\frac{N^2 \log(a_N + 2N)}{a_N} \longrightarrow 0.$$

Then

$$\liminf_{N\to\infty} \frac{\log u_{N,a_N}}{a_N} \ge \frac{s\log 2}{3}, \qquad \frac{u_{N,a_N}}{N^{2r}(N+a_N)^{2r}} \longrightarrow +\infty.$$

In particular, this applies to $a_N = \lfloor N^\beta \rfloor$ for $\beta > 2$, and to exponential shifts.

Proof. The hypothesis implies $a_N/N^2 \to \infty$, since $\log(a_N + 2N) \ge 1$ eventually. In particular $a_N \to \infty$ and $a_N \ge N^2$ eventually. Divide (5) by a_N. The determinant cost tends to zero by hypothesis; the other errors do so because $\log a/a \to 0$ and $\log a/\sqrt{a} \to 0$. This proves the lower limit. Moreover $\log N + \log(N + a_N) = O(\log a_N) = o(a_N)$. Subtracting the logarithm of the denominator in (2) therefore leaves a quantity tending to $+\infty$. □

5 Exact certificates and formalization boundary

The audit checks (3) and (4) by exact rational arithmetic at 16 weight/size/shift choices. It also verifies 28 larger certificates of

$$u_{N,a} \geq 2^b N^{2r}(N+a)^{2r}$$

directly from $T_s(a)^s \geq 2^b D^N N^{2r}(N+a)^{2r}$. For the cubic shift $a = N^3$:

s	$N = 30$	$N = 40$	$N = 60$
3	873	34965	245818
5	1442	58260	409681
7	2011	81556	573545
9	2580	104852	737408

These are exact finite lower certificates, not fitted growth rates.

The Lean growth module verifies power and logarithmic comparison limits and their squeeze applications. The harmonic modules verify both the upper-half and upper-third prime-power divisibilities, their prime-product versions, and the rational denominator budget. The centered determinant estimate and asymptotic large-shift theorem are paper proofs here. Neither the polynomial upper bound for the actual u_{N,a_N} at smaller shifts nor (2) has been proved. The all-odd irrationality result remains outside the conclusions.

References

[1] mathlib contributors, *Chebyshev functions and estimates*, Lean source at the project's pinned revision, theorems `theta_ge` and `theta_le_log4_mul_x`. https://github.com/leanprover-community/mathlib4/blob/85e3a25e006c35636f0e53b0e9296caca2685bc0/Mathlib/NumberTheory/Chebyshev.lean.